

HIKARU OGASAWARA

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PROFILE

Computer Engineering student seeking an internship in infrastructure, security or embedded systems. Delivered projects for Invivio Tecnologia, CPTM and Ibmecc, from firmware to dashboard.

EDUCATION

Ibmecc <i>B.S. in Computer Engineering 2023 – 2027 (expected)</i>	<i>São Paulo, Brazil</i>
Instituto Sidarta <i>High School Diploma 2010 – 2022</i>	<i>São Paulo, Brazil</i>

PROJECT EXPERIENCE

Invivio Tecnologia / Ibmecc — AquaSense, IoT pool monitoring - Developed 887 lines of C++ firmware for the ESP32 monitoring 7 water parameters and switching the solar collector pump on temperature differential, driving an I2C LCD - Implemented MQTT publishing over TLS to a HiveMQ Cloud broker, with consolidated and granular topics, health signals and inbound commands - Built a React and TypeScript dashboard consuming live data over WebSocket, validating payloads with Zod, charting history and falling back to local simulation - Integrated a Portuguese Alexa skill through AWS Lambda and set up GitHub Actions CI with lint, typecheck, tests and build; three-person team	<i>Feb 2026 – Jun 2026</i>
CPTM — Railway asset management system - Developed a REST API in Java with Spring Boot exposing 14 endpoints across 9 controllers to manage trains, stations and lines (creation, retrieval, status updates and removal) - Modeled the operational entities (train, station, line) and implemented persistence through JSON serialization with Jackson - Implemented central tendency calculations (mean, median and mode) over the operational data - Wrote 10 JUnit test classes covering the operations and validated the endpoints with Postman	<i>Feb 2024 – Jul 2024</i>
Ibmecc — Annual executive speaker series website Built the complete website in HTML and CSS for the annual event that brings invited CEOs to share their trajectories, from page structure to published content	<i>Aug 2023 – Dec 2023</i>

TECHNICAL PROJECTS

Enterprise infrastructure and monitoring lab - Provisioned a Windows Server instance from scratch, with Active Directory and Domain Controller, domain, accounts and access policies - Segmented the network and deployed a perimeter firewall with IPFire, defining filtering rules between zones - Deployed a SIEM from scratch in a lab environment, integrating defense layers and event collection for correlation - Virtualized environment nodes on Proxmox and used Kali Linux to validate the applied configurations	<i>Academic</i>
Embedded electronic safe - Ported the same project across three platforms: Arduino UNO, MicroPython on Raspberry Pi Pico and ESP32 - Replaced EEPROM with Preferences/NVS on the ESP32 build and moved the display to I2C LCD (PCF8574), with a 3x4 matrix keypad and servo - Drove the LCD through a library, over I2C and by bit-banging, comparing the abstraction layers	<i>Personal project</i>
Machine learning foundations - Worked through supervised learning in Jupyter with NumPy and pandas: cost-sensitive thresholds, Huber loss, stratified splitting from scratch, learning curves and confusion matrices - Ran an audit that traced a suspicious 0.98 accuracy to temporal data leakage, identifying the offending field	<i>Academic</i>
Python document automation Automated Outlook email draft generation from an Excel spreadsheet with pywin32 and openpyxl, including accent-stripped filename matching for attachments	<i>Personal project</i>

LEADERSHIP AND VOLUNTEERING

Ycare <i>Vice President, since founding 2024 – Present</i> - Coordinates the monthly outreach: on the second Saturday of every month the team delivers food and hygiene baskets to a community - Owns supply collection, basket assembly and volunteer mobilization for each run	<i>São Paulo, Brazil</i>
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SKILLS AND LANGUAGES

Infrastructure: Windows Server (AD/DC), Proxmox, Linux (Ubuntu Server, Pop!_OS), apt, systemd, DNS, DHCP
Security: IPFire, SIEM, Kali Linux, iptables/ufw, network segmentation
Languages: Python, Java, C and C++ (Arduino, ESP32), TypeScript, SQL, MATLAB/Octave
Data and AI: Jupyter, NumPy, pandas, Power BI, advanced Excel, SQL; model evaluation and data leakage detection
Web, APIs and cloud: React, Spring Boot, REST, MQTT, WebSocket, AWS Lambda, Alexa Skills Kit, Postman; HTML and CSS (basic)
Tools and testing: Git, GitHub Actions (CI), JUnit, Vitest, Playwright, technical documentation
Hardware and IoT: ESP32, Arduino, I2C, UART, PWM, LogiSim
Behavioral: Self-motivated, proactive, calm under pressure, team-oriented, communicates across four languages, takes ownership of deadlines
Spoken languages: Portuguese (native); English (fluent reading and listening, working proficiency in speaking and writing); Japanese (intermediate/advanced); Spanish (intermediate)